



GIT AND GITHUB

This workshop offers essential skills for version control and collaborative development.



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A graphic for a 'Git Tutorial' featuring a blue background with a circuit-like pattern. On the right, a white branching diagram with colored dots (cyan, pink, yellow, green) is shown. The text 'Git Tutorial' is prominently displayed in the center-left.

Git
Tutorial



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WHAT IS GIT ?



- Git is a CLI (Command Line Interface) based tool which is used for taking snapshots of each and every version of your code.
 - Git is like an allrounder in a team, If you mean mess up with git is there to cover up.
 - Git can go back in time, can save your work repeatedly many more. It is by far most used tools by the developers.
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VERSION CONTROL SYSTEM

- A version control system (VCS) is a tool that helps manage and track changes to files over time.
 - It is essential for collaborative projects, ensuring that all team members work on the latest version, avoid conflicts, and maintain a history of edits. Whether using tools like Git or cloud-based platforms.
 - VCS provides a streamlined way to handle updates, rollbacks, and collaboration effectively."
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DID YOU KNOW ?



GitHub





GIT INSTALATION AND CONFIGURATION

Configuring user information used across all local repositories

```
git config --global user.name "[firstname lastname]"
```

set a name that is identifiable for credit when review version history

```
git config --global user.email "[valid-email]"
```

set an email address that will be associated with each history marker

```
git config --global color.ui auto
```

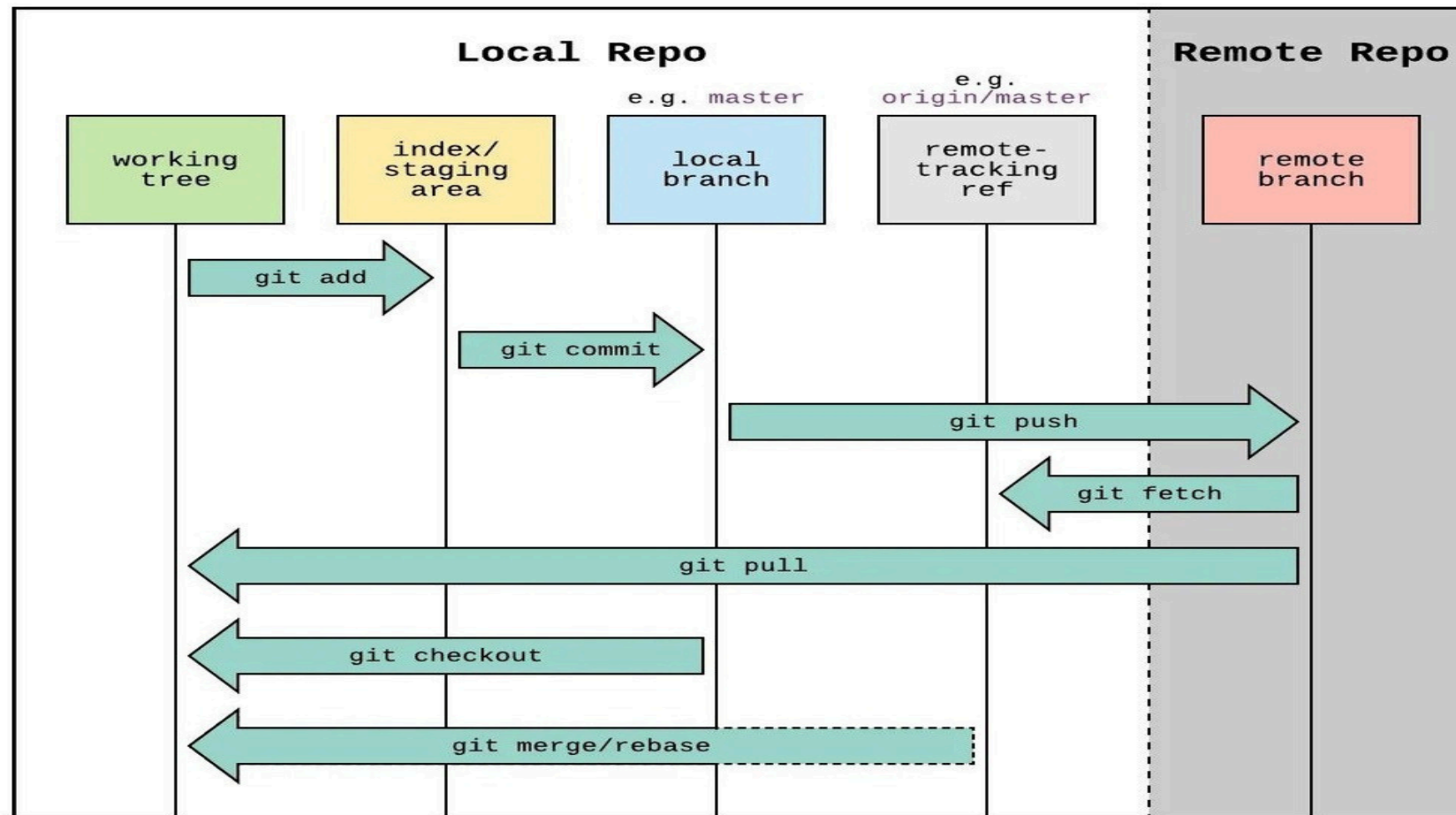
set automatic command line coloring for Git for easy reviewing

WORKFLOW OF GIT



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VERSION CONTROLLING WITH GIT





INITIALIZING GIT PROJECT

- To verify installation, open a terminal and run:

```
bash
```

[Copy code](#)

```
git --version
```

- Run the following command to initialize a Git repository:

```
bash
```

[Copy code](#)

```
git init
```



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ADDING FILES TO VCS

- To track files, add them to the staging area:

```
bash
```

 Copy code

```
git add .
```



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INTRODUCTION TO COMMIT

- A commit is a snapshot of the changes made to the files in the repository at a specific point in time.
- Commits store information about what was changed, who made the change, and when it was made.

Commits on Oct 26, 2024

Add files via upload

 AbhijeetYadav09 authored 3 weeks ago

Verified

ce2684b





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COMMITTING FILES



- Command to Commit Files Individually

bash

 Copy code

```
git commit -m "Your descriptive message here"
```



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STAGGING AREA

- The staging area is a space where Git stores changes before committing them.
 - It allows you to review and control what changes will be part of the next commit.
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LOGGING COMMIT HISTORY

- How you can display and manage commit history in Git

Basic History

bash

 Copy code

```
git log
```

Compact Summary with Graph

bash

 Copy code

```
git log --oneline --graph
```



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GITHUB

- GitHub is a cloud-based platform where you can store, share, and work together with others to write code.
 - GitHub is a platform for hosting code that allows for version control and collaboration.
 - Its name was chosen as a compound of Git and hub.
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REPOSITORIES

- Repo = Repository
 - A folderlike structure containing all your necessary files.
 - Can contain folders, images, files, etc. , anything you want.
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BRANCH

- The branch concept in GIT is like you can create a branch to work on new features or fix issues without disrupting the main code.
 - If your changes in the branch are good , you can merge them into the main code. If not, You can make improvements without affecting the main codebase.
-



PUSH

1. Uploads Local Changes to Remote Repository

- The git push command is used to send local commits and changes to a remote repository.

2. Keeps Remote and Local Repositories in Sync

- It ensures that the remote repository stays updated with the latest local changes, enabling collaboration among team members or syncing your work across different devices.
-



PULL

- **Git pull** is a command which is used to fetch and integrate the changes which are present in the remote repository to the local repository.
It will fetch the changes and merges with the repository.

Syntax

```
git pull [remote-name] [branch-name]
```



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FORK

- A **fork** on GitHub is a copy of an existing repository that allows you to make changes without affecting the original project.
 - Fork is making a copy of a repo.
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PULL REQUESTS (PR)

- A **pull request** (PR) in GitHub is a proposal to merge changes from one branch into another in a repository.
 - Pull requests act as a checkpoint that ensures only thoroughly reviewed and tested code makes its way into the main branch.
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OPEN SOURCE DEVELOPMENT

What is open source ?

- Open source is a term that refers to software that is created and shared with the intention that anyone can: View the source code, Modify the code, and Distribute the code.

Key Features

- Transparency, community-driven development, and freedom to customize.
-

What is GitHub Actions?

- GitHub Actions is a CI/CD (Continuous Integration/Continuous Deployment) tool built into GitHub.
- It helps you automate tasks like testing code, building apps, and deploying.
- You define workflows in `.github/workflows/` directory using `.yaml` files.



Key Concepts of GitHub Actions

- Workflow: Automation process (defined in YAML).
- Event: Triggers workflow (e.g., push, pull_request).
- Job: A set of steps executed on the same runner.
- Step: Individual task in a job.
- Runner: Server that runs your workflows.

When to Use GitHub Actions

- Automatically run tests when code is pushed.
- Build and deploy projects (React, Node.js, Python, etc.).
- Linting, formatting, or checking coding standards.
- Sending notifications, generating reports, etc.

Folder Structure



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```
my-project/  
├── .github/  
│   └── workflows/  
│       └── main.yml  
├── src/  
└── README.md
```

Practical Demo (Steps to Try Yourself)

- Create a GitHub repo and push your project.
- Create folder: `.github/workflows`
- Create a file: `main.yml` or `.yaml`
- Paste the above code.
- Commit → Push → Check Actions tab on GitHub
- Done! Your workflow will run on each push.

What is GitHub Wiki?

A GitHub Wiki is a built-in feature that allows you to create rich, well-structured documentation for your project, directly inside your GitHub repository using Markdown.

Think of it as your project's mini documentation site.

- Wiki is a collection of interlinked pages to store and share documentation.
- GitHub provides Wiki feature for every repository.
- Written in Markdown (.md) format.

Why Use Wiki in GitHub?



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Benefit	Description
Detailed Docs	Separate space for user/developer documentation
Collaboration	Team members can contribute easily
Structured	Multi-page format better than README
Linked	Easily link between wiki pages or to issues/PRs

What is GitHub Projects?

- A Kanban-style task board built into GitHub.
- Used to plan, track, and manage issues, pull requests, and tasks.
- Visual task management — just like Trello or Jira.

Why Use GitHub Projects?



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Feature	Benefit
Visual Planning	Organize tasks in columns
Team Collaboration	Assign tasks to team members
Auto Updates	PR/Issue movement automation
Integration	Linked with Issues and PRs
Agile Friendly	Sprint boards, task tracking

How to Create a Project

- Go to your repository
- Click on Projects tab
- Click New Project → Choose Classic
- Add Columns: To Do, In Progress, Done
- Add Cards: Issues, PRs, Notes
- Assign team members

